

Screw It! Final Report

Your home, your choice – DIY with confidence!

**SCREW
IT!**



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03.04.2026

CMPSC 185

OUR TEAM



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Project Name & Value Proposition

Screw It! is a mobile application that serves as a central hub for home maintenance assistance. Through helping homeowners easily decide between tackling projects themselves through DIY ("Do It Yourself") or hiring professional services, Screw It! aims to simplify home maintenance decision-making. The name "Screw It!" brings down the pressure of doing home maintenance and embodies the nature of doing home maintenance yourself.

The Problem

Renters and homeowners who are new to home maintenance often lack clear guidance on identifying household issues. This guidance is necessary to determine whether they can handle repairs themselves or whether they should hire a professional, as well as how to complete DIY repairs safely and find a reliable expert.

Solution Overview

With a catchy and memorable name, Screw It! uses AI to identify home maintenance issues, compare DIY versus professional options based on cost, risk, and feasibility, and provide step-by-step guidance for completing the repair independently or choosing the right professional.

Needfinding

Interviews – General Practice

During the first phase of development, needfinding, our team interviewed three participants to better understand people's experiences and attitudes towards home maintenance. We had three criteria for finding participants:

1. **Living Situation** – *Do they live in a house or apartment? Would they benefit from learning home maintenance?*
2. **Diversity** – *Gender, culture, lifestyle, occupation, etc.*
3. **Past Experience** – *Have they ever done home maintenance? How much do they know?*

Participants were recruited through personal networks and cold emails to get a range of different perspectives. They volunteered their time and were not financially compensated.

The interviews were semi-structured and used open-ended questions so participants could freely share their experiences. Each interview lasted around 20–30 minutes and was conducted either in person or over the phone. For each interview, we asked a range of questions regarding home and car maintenance. We started by asking about their current living situation and what comes to mind when they think of the phrase "home maintenance." This led into more focused questions like "Can you describe a time when you had difficulty or failed with doing home maintenance by yourself?" We asked participants to walk us through the steps they would take if a theoretical situation arose, such as fixing a clogged shower head. This allowed us to understand what our interviewees needed and to gauge how they feel about home maintenance overall.

Interview #1 – Y.J.

We first interviewed Y.J., a renter with limited knowledge of home maintenance, at the UCSB Library. In her interview, she expressed a lack of confidence in doing home maintenance tasks, stating, "I'm not capable of doing [home maintenance]," and "I would be so happy if someone did [home maintenance] for me." From this interview,

we learned that not everyone feels the need to learn about home maintenance, and people rely on people they know personally rather than calling a professional for help first.

Interview #2 – E.K.

Our second interview was with E.K., an office manager at a home renovation company and wife of the company's owner (a carpenter), over a phone call. Working closely with home renovation projects, she mentioned that "a lot of people who own houses don't know what they don't know." She noted that her time is very valuable, especially as a working mother, and she tends to prioritize efficiency when dealing with home-related issues. One insight from this interview was that she often uses AI (artificial intelligence) tools to guide her through solving home maintenance issues.

Interview #3 – N.N.

Our final interview was with N.N., a homeowner who does basic home maintenance in his spare time, and was conducted in his office. He explained that, "If you don't do it, nobody else does it for you." This motivates him to handle basic home maintenance tasks by himself, but also makes him wish he knew more about home maintenance. However, we noticed that he gave up on learning home maintenance when he didn't have the right tools, suggesting that access to tools and resources can be a blocker in completing home maintenance independently.

Interviews – Synthesis

After each interview, we created empathy maps from our extensive notes taken from what each interviewee had said. This allowed us to see what makes each of their experiences distinct and similar at the same time. From there, we deduced three key learnings:

- **Home maintenance knowledge exists on a spectrum** – Interviewees ranged from a home renovation company employee to a renter with minimal home maintenance experience.
- **Many existing resources are too generic** – Interviewees reported relying on YouTube videos and instruction manuals, but the information they gathered often didn't apply to their own homes.
- **Practice builds confidence** – Interviewees reported a greater willingness to

learn home maintenance after completing a few manageable tasks.

With these findings, we were able to curate “How Might We” (HMW) questions and “Points of View” (POV) to narrow our scope and refine our problem domain to identify what our users need.

Additional Needfinding Interviews

After our initial needfinding interviews, we decided to focus on home maintenance over car maintenance. We interviewed a couple more people to ask more focused questions on this topic.

Interview #1 – B.A.

B.A. is a renter who wants to learn more about home maintenance. B.A. closely represents our target audience because they have limited experience but believe, as they said, “Most of the time it’s better to do it yourself.” The thing they were worried about most with DIY was cost. B.A. didn’t know how much the tools would cost, and was worried that he would just make the problem worse and have to pay more in the end.

Interview #2 – D.D.

Our final interview was with D.D., over Zoom in the library. D.D. is a retired construction worker, handyman, and woodworking teacher. Despite being retired, he still continues to learn new things about home maintenance. The most important thing we learned from this interview was safety. D.D. said, “Safety is important, and it’s always on the top of my mind when I’m working”. He mentioned that safety is especially important for people beginning to learn home maintenance, as they don’t know what’s dangerous.

POVs & EXPERIENCE PROTOTYPES

POV Statements

After the interviews, we focused on three interviewees to better understand their perspectives on home maintenance, and how we might be able to help them with our app.

POV #1 – B.A.

We met B.A., a UCSB student living by himself in a studio apartment he rents.

We were surprised to realize that he thinks it's better to do home maintenance by himself, but ends up using professionals most of the time.

We wonder if this means that he feels uncertain about his own capabilities and, as a renter, prefers to trust professionals to avoid costly mistakes.

It would be game changing to help B.A. feel more comfortable with approaching unknown home maintenance tasks on his own without the fear of damaging his rented apartment.

How might we...

- Reduce the fear of damaging a rented apartment?
- Make renters less reliant on professional services?
- Encourage renters to learn home maintenance skills in a fun way?
- Make it more satisfying and enjoyable to complete maintenance on your own?
- Have friends with the same skills to help instead?
- Make tools more accessible to complete tasks?
- Make home maintenance feel more doable and less intimidating?
- Lessen a renter's financial worries when attempting home maintenance tasks?
- Provide resources to renters to learn how to do it themselves?
- Help renters avoid costly mistakes more often?

POV #2 – E.K.

We met E.K., an office manager at a home renovation company, married to the owner (a carpenter).

We were surprised to realize that she's met many people who are unaware of gaps in their home maintenance knowledge.

We wonder if this means current home maintenance resources are too complex or not designed for someone without prior knowledge.

It would be game changing if homeowners could find simple, easy-to-understand guides that help them identify and fill the gaps in their home maintenance knowledge.

How might we...

- Help homeowners realize the maintenance tasks they're overlooking?
- Help homeowners ask the right questions before problems arise?
- Make obscure issues more visible to the average person?
- Break down home maintenance tasks into actionable steps?
- Translate home maintenance knowledge into simple, understandable language?
- Help homeowners understand urgent vs. minor problems?
- Customize information to fit a specific home?
- Help homeowners feel more confident about learning home maintenance?
- Help homeowners find trustworthy advice?
- Help new homeowners understand what they need to know about maintaining a home?

POV #3 – N.N.

We met NN, a new homeowner who is recently gaining a lot of home improvement experience.

We were surprised to realize that NN enjoys learning about home maintenance, but finds it annoying when a new unexpected problem arises that he isn't prepared for.

We wonder if this means He can only handle problems he's prepared for and relies on professionals otherwise.

It would be game changing if NN could be properly prepared for the problems that could happen.

How might we...

- Make relying on professionals less annoying?
- Make feeling unprepared more manageable?
- Make it enjoyable when an unexpected problem occurs?

- Prepare for the most common problems?
- Teach how to handle unexpected problems?
- Provide the necessary supplies for a variety of problems?
- Make knowing what caused a problem easier?
- Teach what tools to use for what tasks?
- Make relying on professionals cheaper?
- Prevent home maintenance problems from occurring unexpectedly?

TOP 3 Solutions & Experience Prototypes

We focused on three HMWs to explore possible solutions:

- How might we prevent home maintenance problems from occurring unexpectedly?
- How might we help new homeowners understand what they need to know about maintaining a home?
- How might we help renters avoid costly mistakes more often?

From these HMWs, we brainstormed a lot of solutions, and we settled on these three to explore in experience prototypes:

1. An app that ranks the severity of home maintenance issues and says if they should be handled by a professional
2. An app that compares the cost of doing home improvement DIY with professionals
3. An app that tracks the life of a product and predicts when it could malfunction

Experience Prototype #1

The assumption we're testing is that users will be more encouraged to complete maintenance tasks if the app emphasizes how easy they are.

For this prototype, the participant was shown 10 maintenance issues (ranked by severity) and asked if they would do it themselves, and created 2 piles based if it's doable or not to them. The participant did it again but was given information and statistics about whether it's a common and easy DIY, and if they should call professional services. They then created 2 piles based on if it's doable or not to them.

From this experience prototype we concluded that our assumption was valid. The participant felt more confident in their ability to do a task if they knew many other people did them.

Experience Prototype #2

The assumption we're testing is cost is the primary driver in the DIY vs. professional decision.

For this prototype, the participant was presented with both a verbal and pictorial description of six different maintenance issues. The participant was then given tables for each issue containing the cost, time, and risk for DIY vs. professional repair. The participant was asked to choose between DIY vs. professional repair and rank cost, time, and risk according to their influence on the decision.

From this experience prototype we concluded that our assumption was valid. Cost was often ranked the highest among the factors influencing the participant's decision. However, when risk was high, that was more important than cost.

Experience Prototype #3

The assumption we're testing is that people want to know when something could break before it happens.

For this prototype, the participants were given things that would break while distracted doing other things. They were given increasingly specific times about when and how the things would break. When things broke, they were tasked to fix it. At the end, they were asked how they felt.

From this experience prototype, we concluded that our assumption was not valid. Participants said that knowing when the thing would break did not remove the stress of having to fix it.

DESIGN EVOLUTION

Final Solution

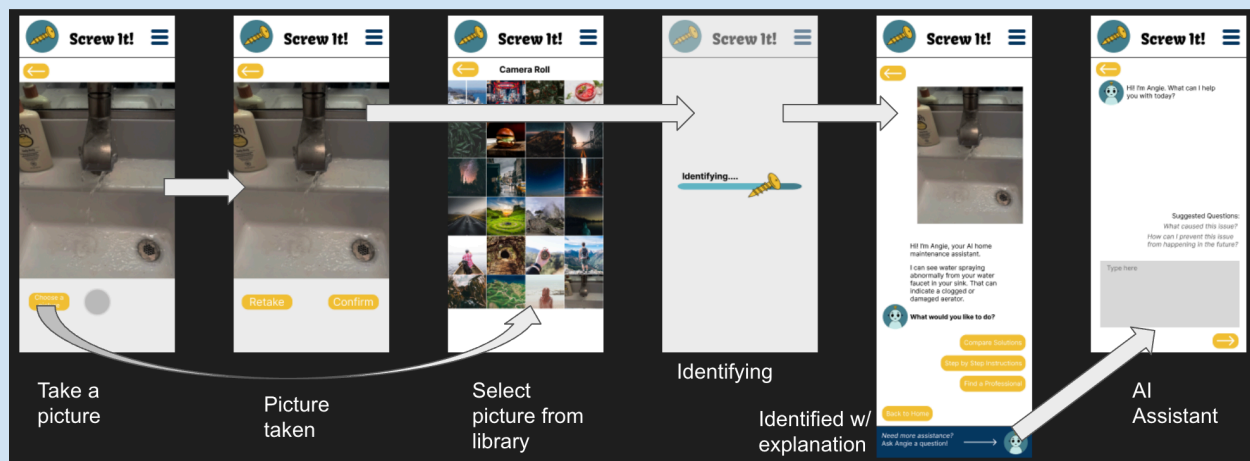
Comparing the impact, reach, and novelty of the three solutions we did experience prototypes for, we decided to combine our first and second prototypes. The last prototype didn't match our assumption at all, so we ignored it. However, the other two prototypes both seemed to work, and since they were both related to ranking and comparing issues, we decided to combine them. As such we decided on this solution: An app that ranks home maintenance tasks by how common it is to DIY, how risky it is to DIY, and how much it costs to DIY compared to having a professional do it.

Tasks

Once we had an idea for an app, we had to split up the app into a series of tasks we wanted the app to be able to help the user accomplish. These tasks were simple, moderate, and complex.

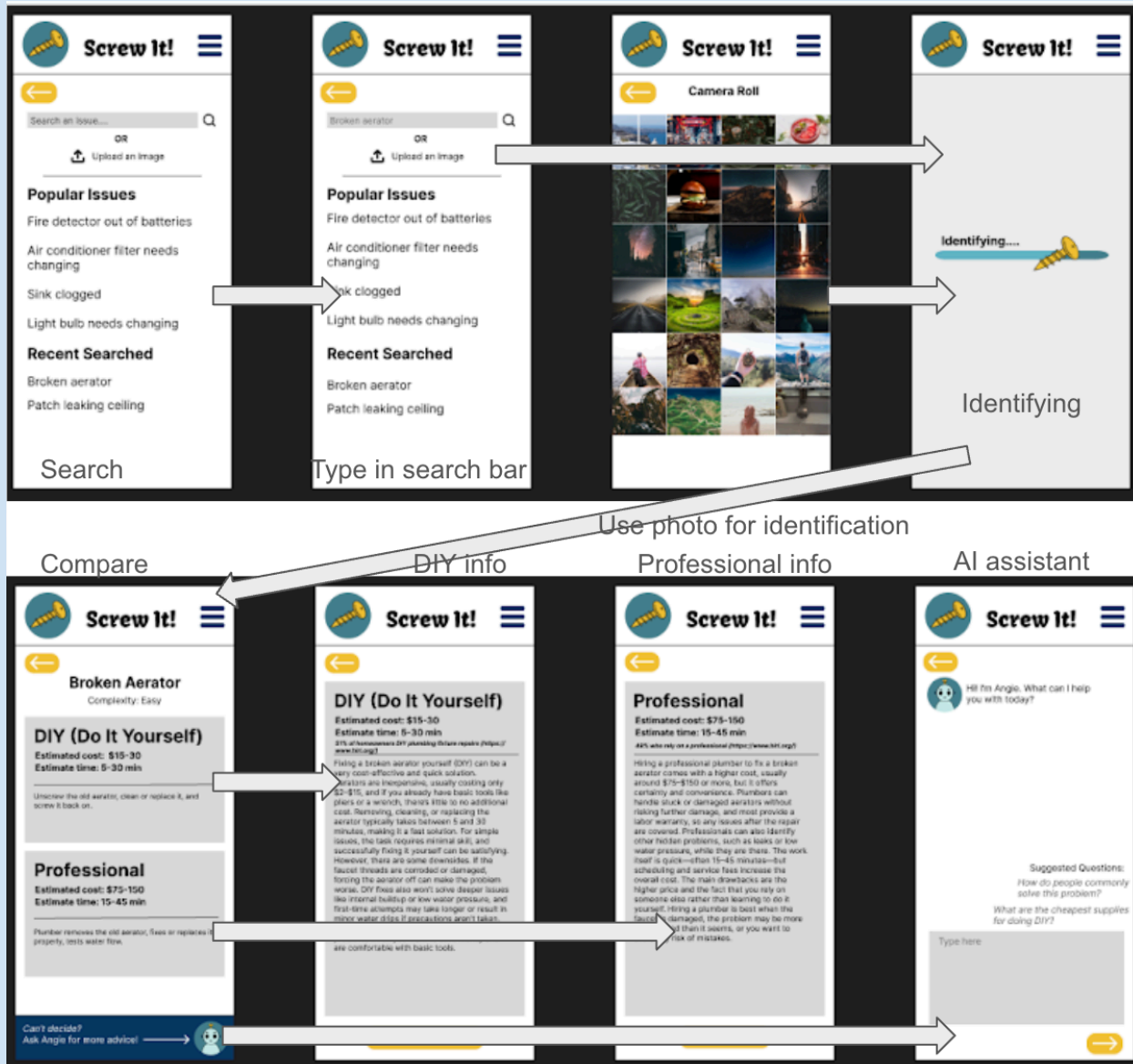
Simple Task: The user can upload or take a picture to identify the maintenance issue

Our simple task was to have the user identify the issue they want to solve. This is important for the rest of the app's features. All that the user is doing is taking or uploading a picture, so it was considered simple.



Moderate Task: The user can compare the DIY and professional solutions

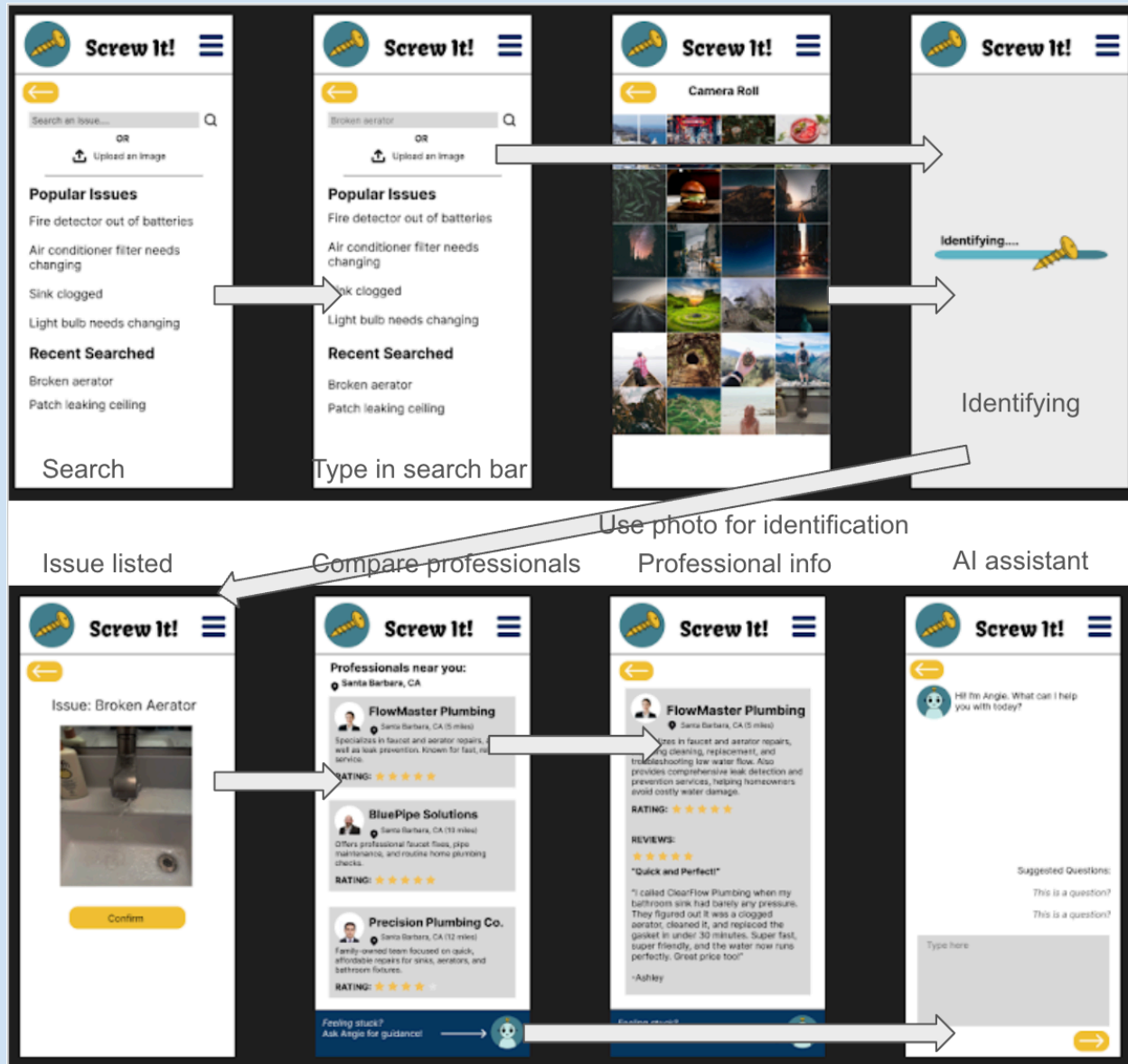
Our moderate task was to have the user compare information between the DIY solution and a professional solution, such as time, cost, and risk. This is one of the key features of our app, connected directly to our solution. This requires mental effort of the user, so was considered moderate.



Bonus Moderate Task: The user can find and book a professional

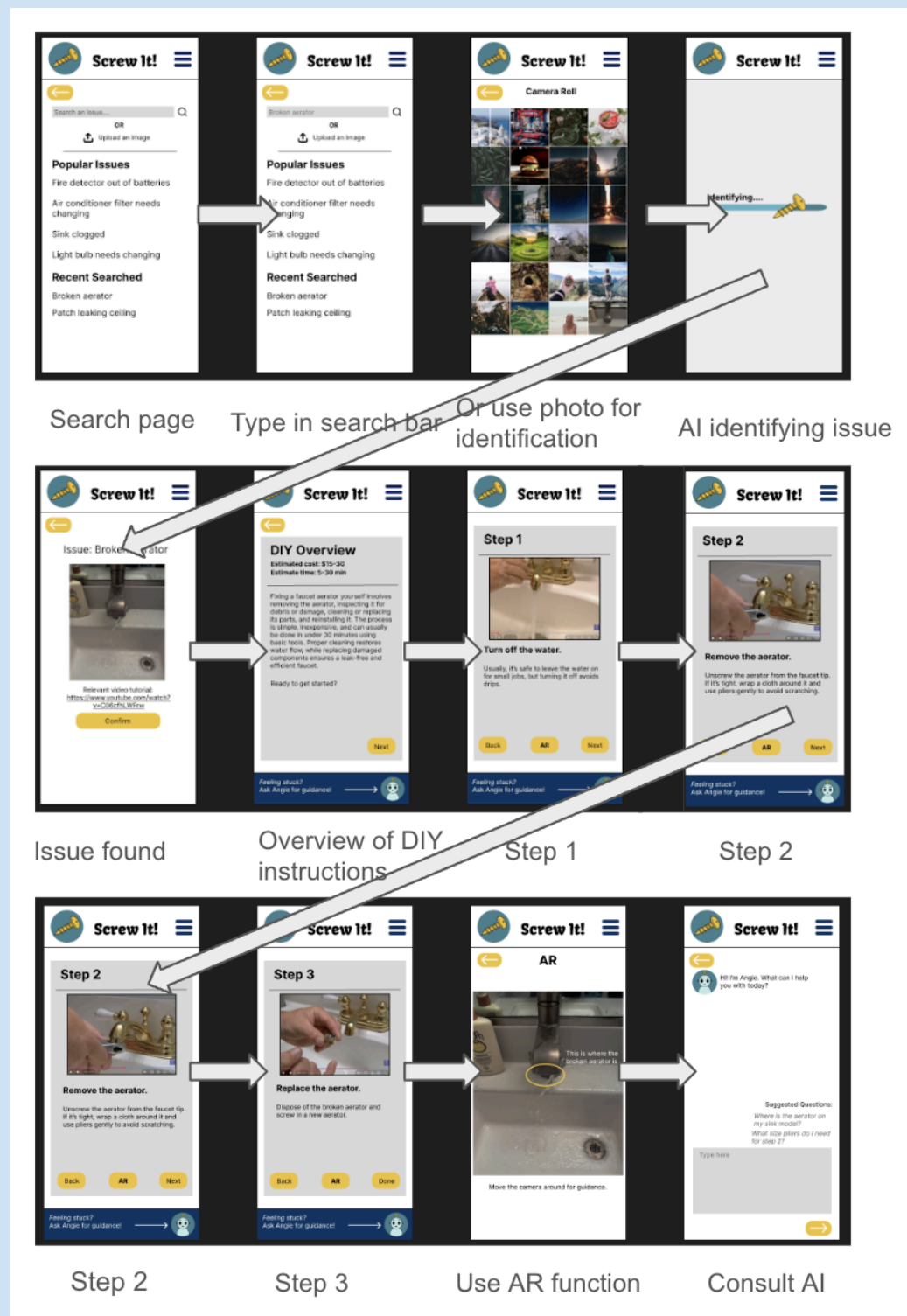
We added this task later to expand on our app's professional solution assistance.

The user can compare different professionals nearby to find the best fit for solving the issue, and then book the professional directly from the app. This task has similar complexity to the other moderate task, so it was considered moderate.



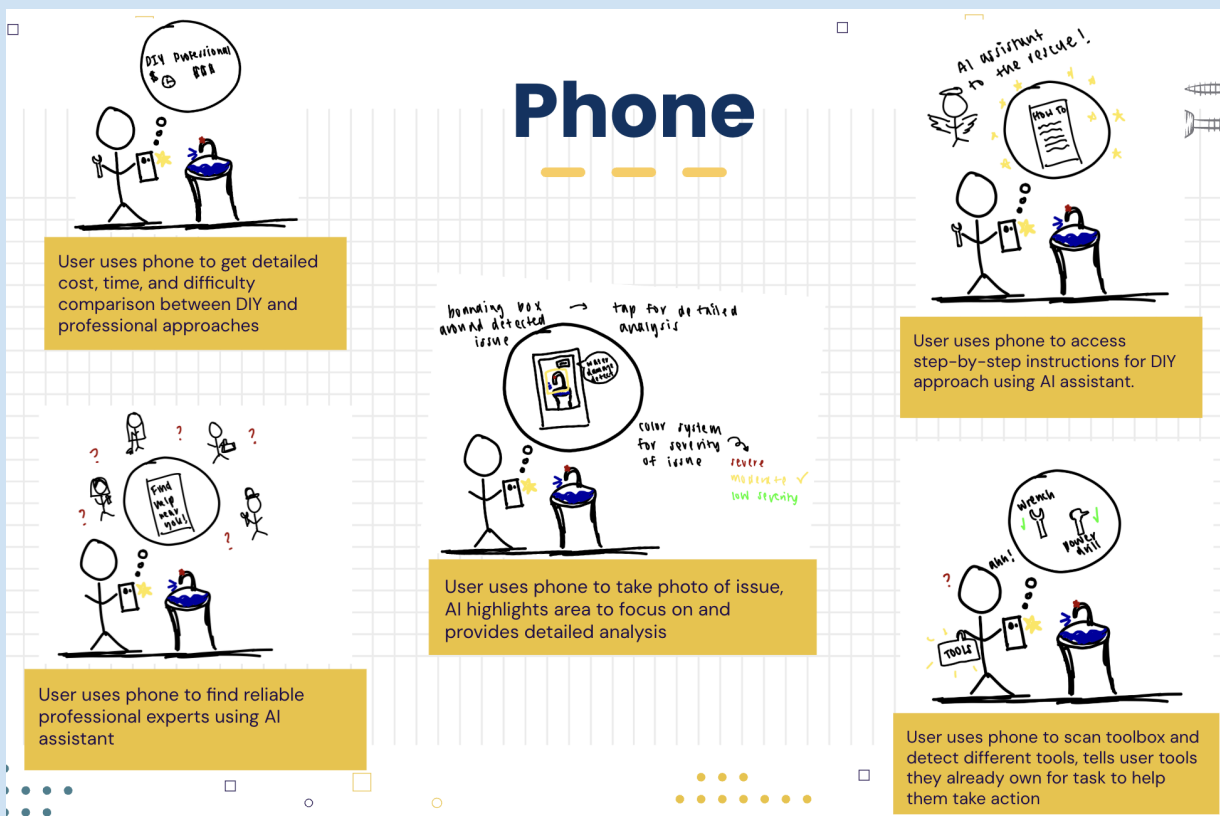
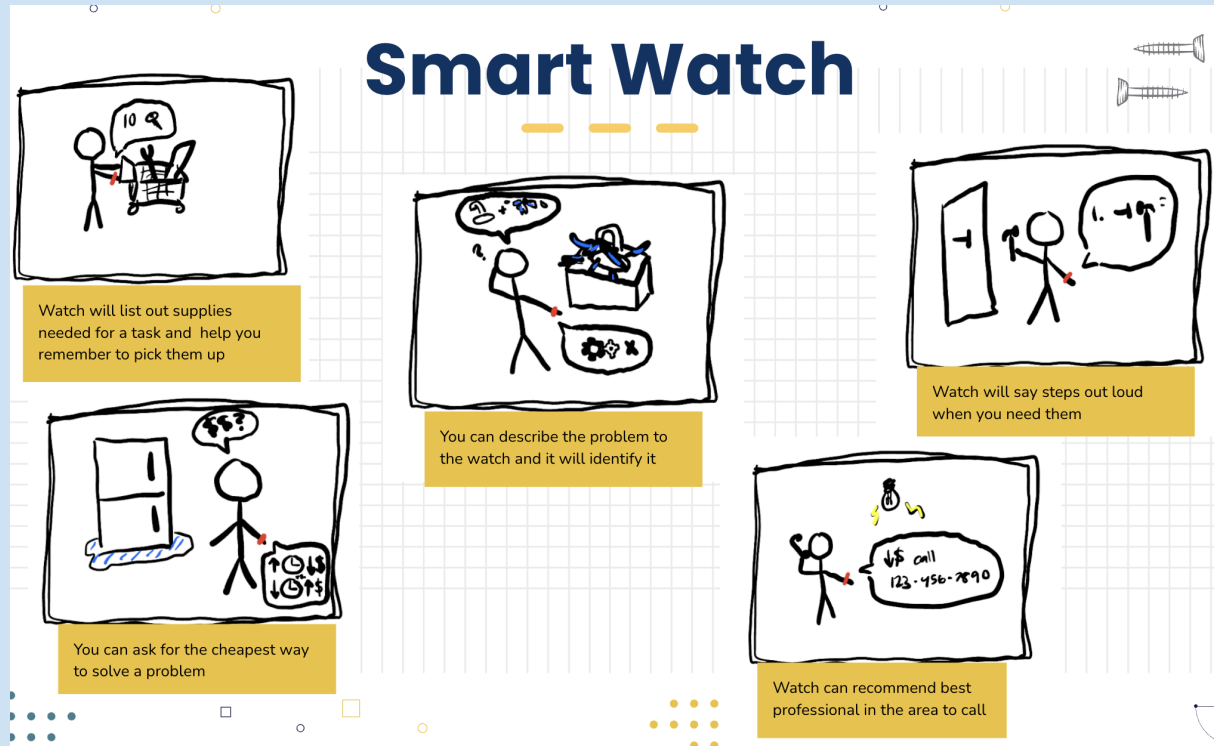
Complex Task: The user can learn how to fix the home maintenance issue

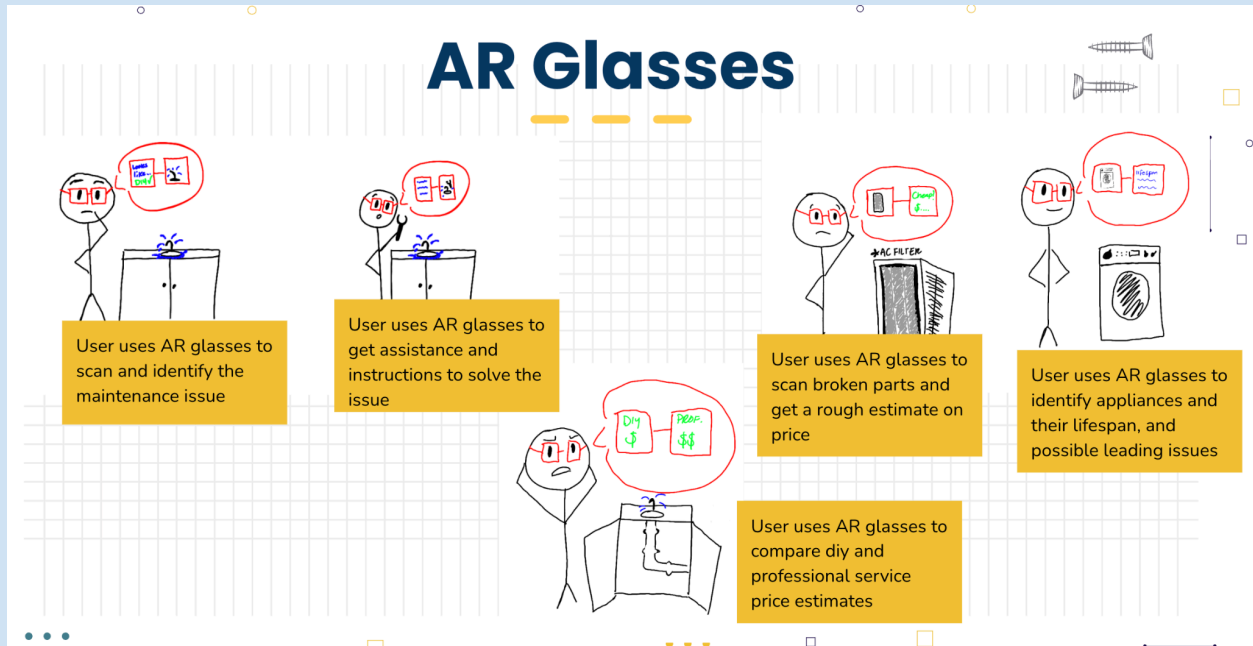
Our complex task is guiding a user through a DIY repair, which requires a lot of effort from the user end. With a variety of features to help the user learn and understand the steps, that also means a variety of features the user needs to use. As such, this is a complex task.



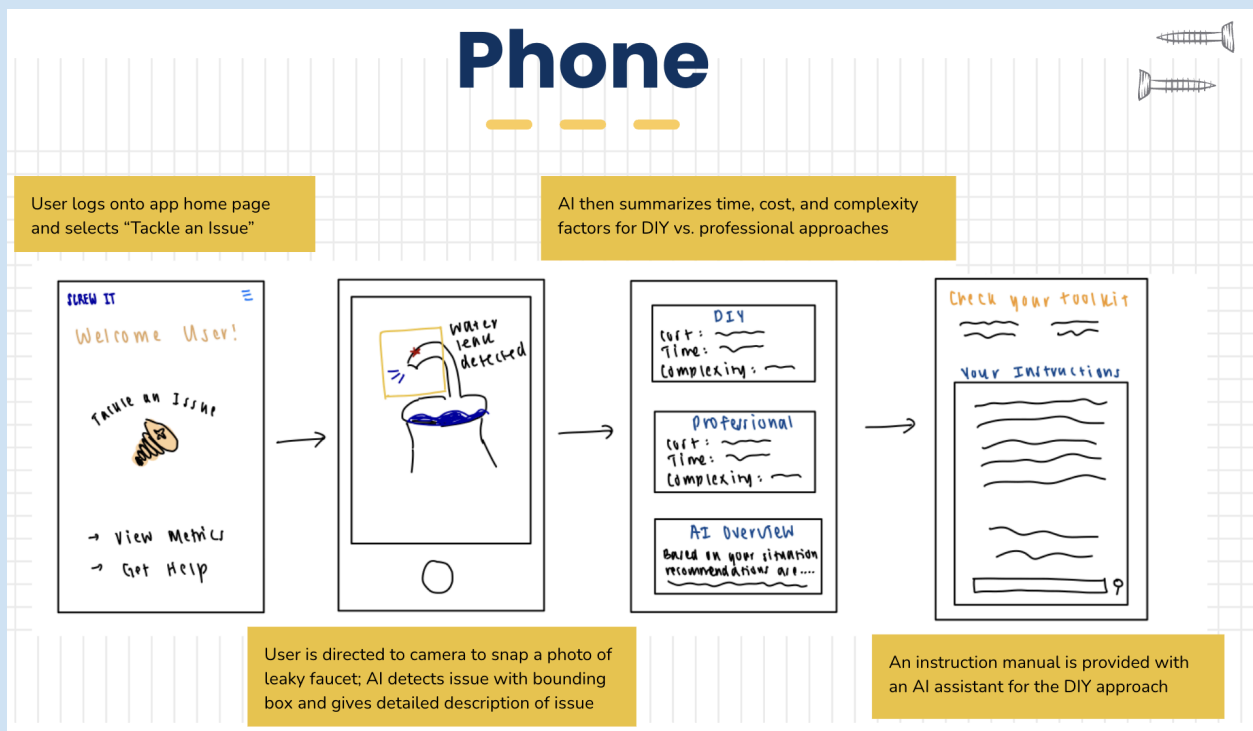
Design Over Time

Our design had many changes over time. First, we started by exploring different modalities to see which best fit the solution to our problem.

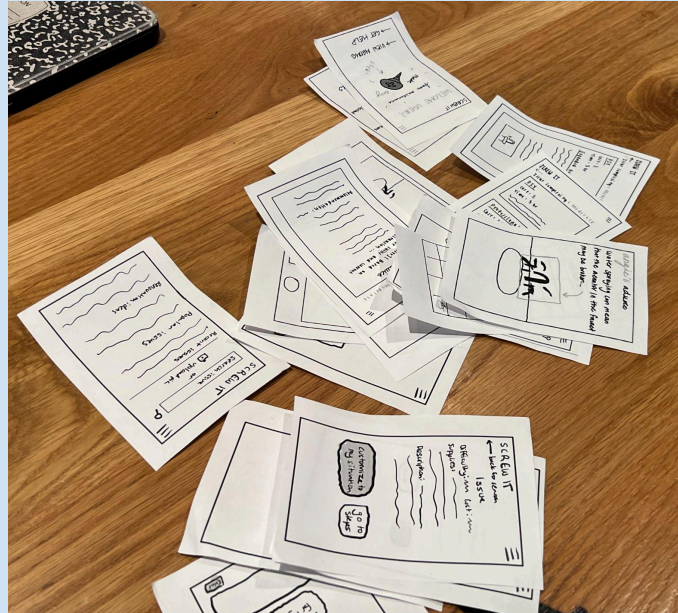
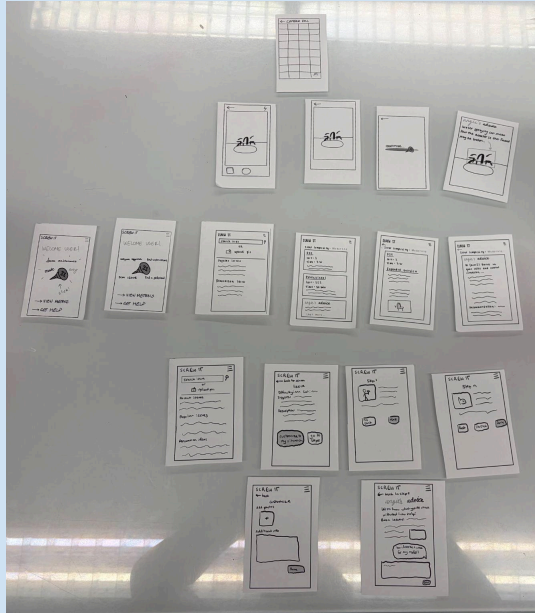




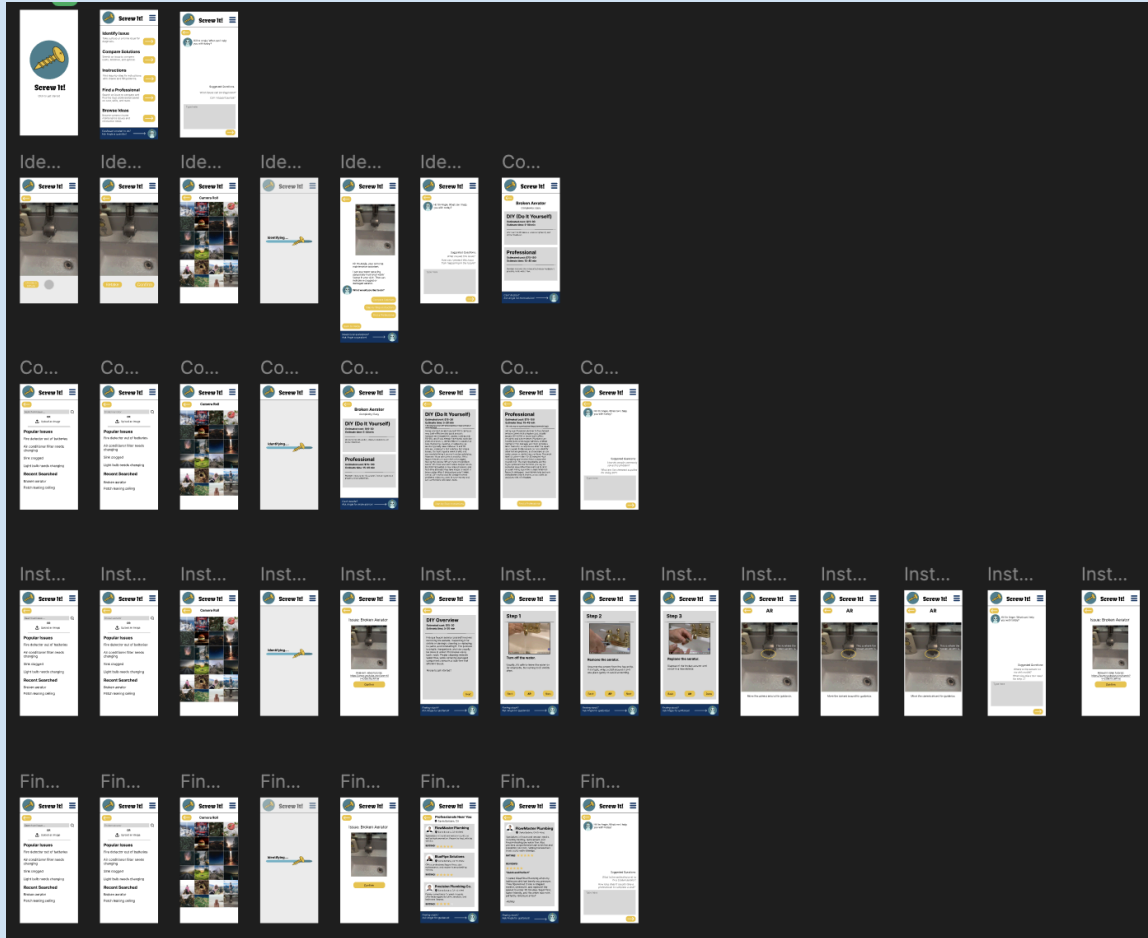
We chose the phone modality, as this allowed us to reach the largest user base while implementing most of our desired features, even if a more hands off option could have worked better for some tasks.



To evaluate this design, we made a paper prototype. We then had people test it out, looking for how effective it was in terms of intuitive design, helpfulness, and ability to help you learn.



We tested the prototype with volunteers at a Starbucks. The participants often struggled to navigate to the correct task, so our design needed improvement to become more intuitive. Many also struggled to find the AI assistant meant to help them. However, all the participants ranked the app as being very helpful for the tasks they were completing. The participants also thought its ability to help you learn was high, although one person mentioned the importance of videos for learning home maintenance. From this advice, we reworked the design of our home page, made finding the AI assistant easier, and reworked the complex task to include video as well as AR assistance. With these changes in mind, we implemented a med-fi prototype in Figma, to better mimic the experience of using an app.



We had outsiders with no knowledge of our project evaluate our design and find issues according to the 10 usability heuristics of design. Here is a comprehensive list of the severity 3 and 4 heuristic violations we received, and how we addressed them, or why we did not.

Severity 4

<u>Problem</u>	<u>Heuristic</u>	<u>Description</u>	<u>Solution</u>	<u>Fixed?</u>
1	H1	AI assistant gives no feedback when clicked	Even in prototype form, simulate a basic AI response flow or displace placeholder conversation	Yes, AI responses were simulated.

			feedback	
2	H5	No confirmation before committing to a professional hire	Add a confirmation page summarizing cost, service, and date before booking	Yes, we added a way to book a professional hire through the app.
3	H9	AI provides no recovery suggestions	Simulate contextual Q&A responses	Yes, AI responses were simulated.
4	H3	On the DIY tab, the user is not able to access the step-by-step instructions	Map the button to the correct instructions, rather than it being a dummy button	We could not replicate this issue, but made sure all flows were correctly connected.
5	H9	AI chat assumes that the user's uploaded picture has enough context to analyze	Design a clear error message from the AI, Angie, and allow users to retake photos	Yes, we added confirmation from the AI that the photo was clear enough.
6	H7	The instructions flow forces the user to view one step at a time and manually click "next."	Include a "View all steps" button or an overview table of contents for the user to select a specific step	Yes, an overview of steps was added to the instructions
7	H5	The user can accidentally click the top of the	Allow the user to access the most recent page from the	We didn't agree with this one. It's important to be able to navigate

		screen, which redirects the app back to the home page	home page if they were in the middle of a DIY	back to the home screen, and it is not that hard to recover your position if you misclick.
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Severity 3

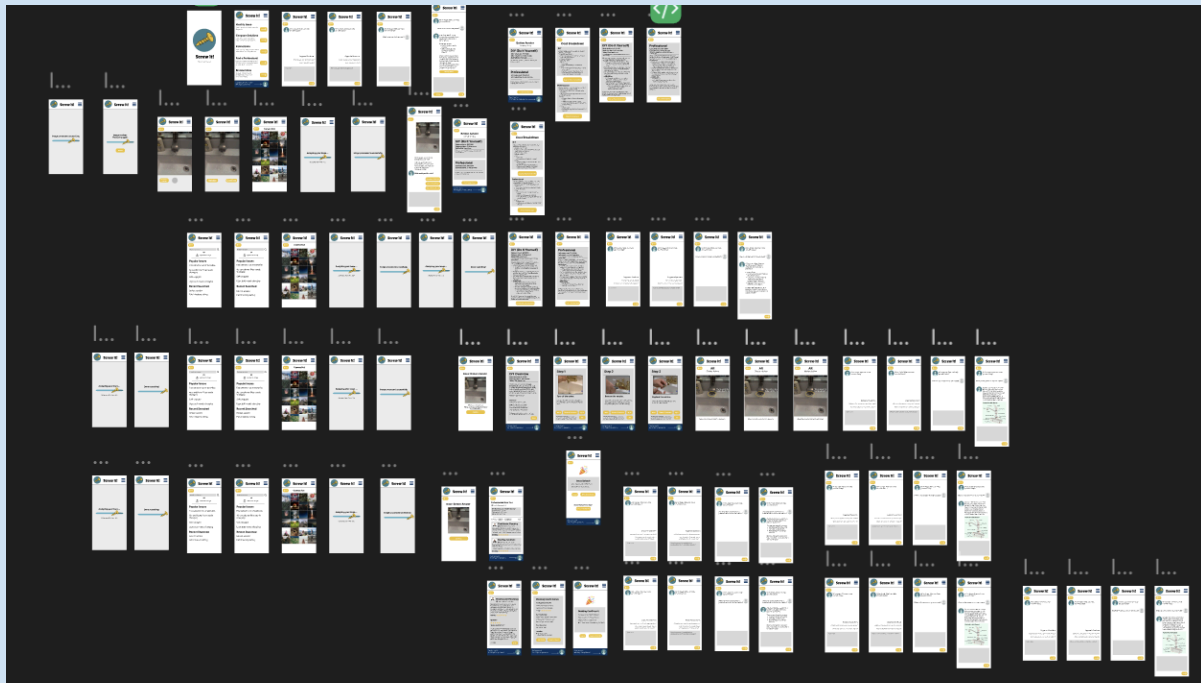
<u>Problem</u>	<u>Heuristic</u>	<u>Description</u>	<u>Solution</u>	<u>Fixed?</u>
1	H1	No clear loading state when image is uploaded	Add animated loading indicator with message like "Analyzing your issue..." and estimated wait time	Yes, added wait time estimate to loading.
2	H1	AR mode lacks status feedback	Add a persistent "AR Mode Active" banner with exit button	Yes, added "Status: Active" to AR screens
3	H2	Cost comparisons lacks real world breakdown	Provide expandable cost breakdown sections	Yes, provided a detailed and expanded cost breakdown in the comparison section.
4	H4	Inconsistent AI button placement	Standardize AI floating button placement	We could not figure out what this meant, as we standardized the AI button to always appear at the bottom of

				the screen when available.
5	H5	No warning about DIY risk level	“Safety Risk Level” indicator with caution messaging	Yes, added risk to comparison.
6	H6	Users must remember issue details across screens	Keep issue summary card pinned at top	Partially yes, we added this to some areas, such as the professional comparison, but felt it was unnecessary everywhere.
7	H7	No filtering for professional search	Add filter and sorting controls	Yes, added visuals for filtering.
8	H9	No clear error message if image unclear	Add message like “image unclear - try better lighting”	Yes, addressed earlier.
9	H10	No onboarding tutorial	Add onboarding walkthrough	We did not address this issue. We felt our design was intuitive enough to understand on its own, although we could add this feature in the future.

10	H4	After uploading the image to Angie, the three buttons given are “Compare Solutions”, “Step by Step Instructions,” and “Find a Professional.”	Remove the “Find a Professional” button	We strongly disagreed with this. “Find a Professional” is one of the features of our app, so it doesn’t make sense to remove it.
11	H6	When viewing the list of professionals for the “broken aerator,” the cost estimate and time commitment data from the “DIY” tab are no longer visible	Maintain a little section when clicking on either the DIY or professional tab with the other’s cost + time for easy comparison	Yes, cost and time were added to professional comparison.
12	H5	After confirming the image being uploaded, the AI assumes that the image is able to be processed successfully	Have a screen or indication that shows Angie successfully processed the image to confirm to the user	Yes, addressed earlier.
13	H5	The app allows users to manually type their issues into the search bar but doesn’t have any fallback for typos or predictive	Implement a predictive search dropdown that suggests correct terminology as the user types	We didn’t implement this. This issue is a limitation of figma, and not our app design.

		text/auto-completeness		
14	H5	When uploading an image, there is no fallback for cases with failed image uploads	Add a reupload button or have an indicator that says the photo could not be processed	Yes, addressed earlier.

Finally, with all of these suggestions, we implemented our high-fi prototype in Figma, updating our med-fi design with these changes.



FINAL PROTOTYPE

Our final prototype was implemented using Figma, which helped us to prototype our features without having to make sure they function. It allowed us to mimic our app and task flows without spending the amount of time it would take to actually create it.

Figma Pros:

- Prototyping allows us to mimic the flow of tasks
- Easy to make a nice design
- Easy collaboration

Figma Cons:

- Tricky to learn
- Requires hard-coding

Wizard of Oz & Hard-Coded Features:

- The issue is always with a broken aerator.
- The video shows screenshots instead of video playback.
- AR is just a photo and cannot show moving around and 3D features.
- AI's responses are added by us and aren't exactly accurate.
- The professional services and reviews are not real and made up.

Additional AI Tools Used:

Claude was used to help with the website building, fixing the functionality, and adding additional aesthetic features to ensure that we showcase the entire journey of our project.

REFLECTION & NEXT STEPS

Reflection

One of the main things we learned this quarter was the importance of testing. Through needfinding, testing, and prototyping, what we thought would be the right problem, the right solution, the right design, ended up being different from what the evidence showed. By keeping an open mind to what our testing was showing us, rather than sticking to what we thought would be right, we were able to create a better app. In the same line of thinking, we also learned the importance of iterating. You can't come up with a perfect solution on the first try, so it's important to keep learning and keep adjusting.

One of the skills we improved the most in was brainstorming. We hadn't originally thought of it as a skill, but constantly having to contract and expand our ideas stretched brainstorming muscles we didn't know existed. This is a skill that will be useful for whatever we decide to do in the future, whether it's design related or not.

Next Steps

If we had more time just for the design process, we would add more features. We had an idea for browsing popular home renovation ideas, to give users motivation and inspiration for improving home maintenance skills not only when there's an issue, but also for renovation in general. Additionally, we would finetune the design implementation of our simple task, identifying an issue with a photo. Although simplified for the sake of design, a single photo would likely be unable to identify an issue in reality. We would do more research, possibly with additional needfinding with professionals, to figure out what information and how much is needed to be confident in identifying an issue. Despite these things, we are very happy with the current state of our app design and feel it captures much more than what we anticipated going into the project.